MILITARY SPECIFICATION

TUBING, ELECTRICAL INSULATION EXTRUDED POLYTETRAFLUOROETHYLENE

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of flexible tubing entitled Tubing, Electrical Insulation Extruded Polytetrafluoroethylene, which is used with the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-406 - Plastics, Organic: General Specifications Test Methods.

MILITARY

MIL-M-14077 - Molding, Plastic, Polytetrafluoroethylene (TFE-Fluorocarbon Resin).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

FSC NUOR

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 The following documents, of the issue in effect on date of invitation for bids, form a part of this purchase description.

AMERICAN SOCIETY FOR TESTING MATERIALS ASTM D 14557-56T.

Tetraflouroethylene Resin, Molding and Extrusion Materials ASTM-D-876-54T.

(Copies may be purchased directly from the American Society for Testing Materials 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the tubing shall be made from material conforming to MIL-M-14077 or equal. The contractor shall report to the procuring agency the trade name of the material used to fabricate each lot. Unless otherwise specified, the contractor as fabricator, may select the class of material to be used in producing the tubing.

3.1.1 Color. Unless otherwise specified, the tubing shall be natural in color, ranging from translucent white to opaque white.

3.1.2 Finish. Unless otherwise specified by the procuring activity, the tubing shall have the natural, reasonably smooth, finish characteristic of polytetraflouroethylene when cooled from the sintering temperature without restraint. When a smooth finish is specified, no machining or buffing of surfaces shall be permitted except to remove flash, burrs or projecting ridges.

3.2 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.3 Dimensions and tolerances. The dimensions and tolerances of the tubing shall be in accordance with Table I as specified by the procuring activity, (see 6.1).

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Table I

Standard Sizes and Tolerances*

Size No.	Inside Diameter, Inches			Wall Thickness, Inch	
	Nominal	Minimum	Maximum	Nominal	Tolerance Plus or Minus
24	0.022	0.020	0.026	0.012	0.003
23	0.026	0.023	0.030	0.012	0.003
22	0.028	0.026	0.032	0.012	0.003
21	0.032	0.029	0.036	0.012	0.003
20	0.034	0.032	0.039	0.016	0.003
19	0.038	0.036	0.044	0.016	0.003
18	0.042	0.040	0.049	0.016	0.003
17	0.047	0.045	0.054	0.016	0.003
16	0.053	0.051	0.041	0.016	0.003
15	0.059	0.057	0.067	0.016	0.003
14	0.066	0.064	0.072	0.016	0.003
13	0.076	0.072	0.082	0.016	0.003
12	0.085	0.081	0.089	0.016	0.003
11	0.095	0.091	0.101	0.016	0.003
10	0.106	0.102	0.112	0.016	0.003
9	0.118	0.114	0.124	0.020	0.004
1/8 in.	0.125	0.120	0.130	0.020	0.004
8	0.133	0.129	0.141	0.020	0.004
7	0.148	0.144	0.158	0.020	0.004
6	0.166	0.162	0.178	0.020	0.004
5	0.186	0.182	0.198	0.020	0.004
4	0.208	0.204	0.224	0.020	0.004
3	0.234	0.229	0.249	0.020	0.004
1/4 in.	0.255	0.250	0.260	0.020	0.004
2	0.263	0.258	0.275	0.020	0.004
1	0.294	0.289	0.311	0.020	0.004
5/16 in.	0.321	0.313	0.329	0.020	0.004
0	0.330	0.325	0.347	0.020	0.004
3/8 in.	0.387	0.375	0.339	0.025	0.005
7/16 in.	0.451	0.438	0.462	0.025	0.005
1/2 in.	0.515	0.500	0.524	0.025	0.005
5/8 in.	0.643	0.625	0.655	0.030	0.006
3/4 in.	0.772	0.750	0.786	0.030	0.006
7/8 in.	0.901	0.875	0.911	0.035	0.007
1 in.	1.030	1.000	1.036	0.035	0.007
1-1/4 in.	1.287	1.250	1.290	0.040	0.007

*Unless otherwise specified, the listed sizes are standard, and the tolerances apply at 75-85°F.

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3.4 Properties. The tubing properties shall conform with Table II when tested as specified in the applicable paragraph.

Table II

Property	Value	Applicable Paragraph
Melting point degrees Fahrenheit (°F)	620 ± 18°F	4.4.1
Specific Gravity	2.14 - 2.20	4.4.2
Tensile Strength at 77 ± 2°F pounds per square inch (psi)	2000	4.4.3
Elongation at 77 ± 2°F percent	100-300	4.4.4
Stress Relief (shrinkage) at 350 ± 2°F percent, maximum (max)	2.0	4.4.5
Heat Aging (weight loss) after 3 hours at 572 ± 2°F percent, max	0.05	4.4.6
Dielectric Strength at 77 ± 2°F volts per mil, minimum (min)		4.4.7
Wall Thickness, in.		
0.010 to 0.020 incl.*	900	
over 0.020 to 0.035 incl.	500	
over 0.035 to 0.060 incl.	400	

*inclusive

3.5 Workmanship.

3.5.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.5.3 Burrs. Parts shall be free from burrs.

3.5.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.5.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit twenty-five consecutively produced feet of tubing for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

Table III

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Melting point (see 4.4.1)	X	
3. Specific gravity (see 4.4.2)		X
4. Tensile strength (see 4.4.3)	X	
5. Elongation (see 4.4.4)		X
6. Stress relief (see 4.4.5)		X
7. Heat aging (see 4.4.6)		X
8. Dielectric strength (see 4.4.7)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Tubing.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Dimensions and tolerances (see 3.3)	Approved Gage
	102. Finish improper	Visual
	103. Tubing warped, twisted, etc	Visual
	104. Evidence of poor workmanship (see 3.5)	Visual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.5)	Visual

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4.3.3 Testing. All tests shall be performed. Tests shall not be performed in the order specified below:

Test	Test Classification
a. Melting point	Major *
b. Specific gravity	Major *
c. Tensile strength	Major *
d. Elongation	Major *
e. Stress relief	Major *
f. Heat aging	Major *
g. Dielectric strength	Major *

*Acceptable Quality Level (AQL) 0.10 percent.

Of these tests, a and c are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Melting point and tensile strength. Three pieces of tubing, each approximately one foot long, making a composite sample representative of the lot, shall be selected from each lot. The sample shall be subjected first to the tensile strength test (4.4.3) and then to the melting point test (4.4.1). If any item fails to comply with 3.4, the lot shall be rejected.

4.3.3.2 Specific gravity, elongation, stress relief, dielectric strength and heat aging. Three pieces of tubing, each approximately one foot long, making a composite sample representative of the lot, shall be selected from each lot. The sample shall be tested in accordance with 4.4.2, 4.4.4, 4.4.5, 4.4.7 and 4.4.6 at an AQL of 0.10 percent. If any item fails to comply with 3.4, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major and Minor characteristics and tests. Inspection and testing shall be performed with the acceptance test equipment, operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Melting point.

4.4.1.1 Preparation of specimen. Three specimens shall be prepared as shavings 10 to 15 mils (0.001 inch) thick, 50 to 70 mils wide and 1/2 to 3/4 inch long from each sample. Conditioning will not be required.

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4.4.1.2 Procedure. The specimen shall be tested in accordance with ASTM D 1457-56T.

4.4.2 Specific gravity.

4.4.2.1 Preparation of specimen. Five sample units shall be selected and conditioned for a minimum of 16 hours at a temperature of 73.4 ± 2°F.

4.4.2.2 Procedure. The specimens shall be tested in accordance with method 5011 of L-P-406, modified if necessary, by using a balance or scale with a larger capacity than an analytical balance and by adding a few drops of wetting agent to the water. The determination shall be made with an accuracy of ± 0.5 percent. If the first set of five parts fails to meet the requirement an additional five parts shall be tested and the results of the ten tests shall conform to the requirement. If the parts are uniform in weight within ± one percent, the specific gravities may be assured to be uniform ± one percent and weight measurements may be substituted for specific gravity determinations.

4.4.3 Tensile strength. Three specimens of sufficient length shall be selected and tested at a temperature of 77 ± 2°F, in accordance with ASTM D 876-54T.

4.4.4 Elongation. Three specimens shall be selected and tested at a temperature of 77 ± 2°F, in accordance with ASTM D 876-54T.

4.4.5 Stress relief.

4.4.5.1 Preparation of specimen. Three specimens shall be selected and conditioned at a temperature of 73.4 ± 2°F for a minimum of 16 hours.

4.4.5.2 Procedure. The inside diameter and wall thickness of the specimen shall be measured with an accuracy of ± 0.0001 inch and heated to 350 ± 2°F for 1 ± 0.1 hours for each 1/4 inch of thickness. Subsequently, the specimens shall be cooled at a rate of 82.4 ± 5°F per hour until the outside has reached 212°F and then subjected to free cooling, protected from strong drafts, until at least one hour after the outside of the tubing has cooled to 73.4 ± 2°F. The dimensions shall be remeasured at the original location and the percent change calculated as follows:

$$\text{Percent change} = \frac{(100)(\text{change in dimension})}{(\text{original dimension})}$$

4.4.6 Heat aging. Three specimens shall be selected and tested at a temperature of 572 ± 2°F as specified in ASTM D876-54T, method B.

4.4.7 Dielectric strength. The dielectric strength of the tubing shall be determined in accordance with ASTM D876-54T at a temperature of 77 ± 2°F.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The tubing shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The tubing shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The tubing shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Intended use. Polytetraflouroethylene flexible tubing procured under this purchase description is intended primarily for use as electrical insulating tubing capable of withstanding temperatures up to 500°F.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-2138 dated 30 October 1960.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A050

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

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